
Human Performance Testing

A final report submitted to:

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EXECUTIVE SUMMARY

The project focused on evaluating human muscle power characteristics through the development of two systems: the bike dynamometer and the hand dynamometer. The goal was to experimentally determine the torque vs velocity and force vs velocity relationship of different muscle groups and compare the results to the expected theoretical behavior expected.

The bike dynamometer was improved to measure the brake torque applied by the cyclist as opposed to using the input torque to reduce the error caused by the flywheel's inertia. The hand dynamometer was refined from previous design by incorporating a stronger locking mechanism, and improved testing methodology to get more consistent and accurate runs. The hand dynamometer experimental testing was conducted across multiple resistance levels, and data collected and processed using LabVIEW. The resulting force vs velocity curves showed agreement with theoretical expectations, confirming the concave up response curve for human muscles power characteristics. Some sources of error were identified, including user flinching from the release mechanism, inconsistent forces applied across a run, the effects of fatigue during the higher resistance runs. However, despite these challenges the system was able to capture the expected overall trends for human muscle behavior. For future work we will be building and starting data acquisition on the bike dynamometer so we have the muscle characteristics for the leg and arm muscles to further prove our theoretical behavior and refine our data processing methods and increase the number of test subjects to strengthen the reliability of results.

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1 INTRODUCTION

1.1 Background

When looking at the power characteristics for torque vs speed for an electrical system see get a response curve that is linear as seen in Figure 1.

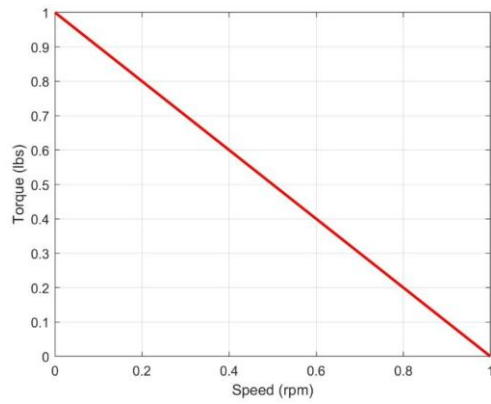


Figure 1. Electrical system power characteristics

A hydraulic system has a power characteristic for torque vs speed has as a concave down response curve as seen in Figure 2.

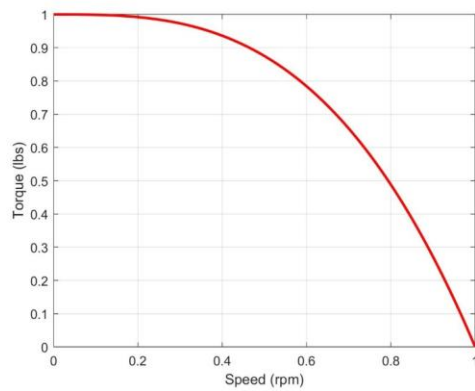


Figure 2. Hydraulic system power characteristic

Finally, when looking at a human being's muscle characteristics our theory is that the response curve for torque vs speed will be concave up as seen in Figure 3.

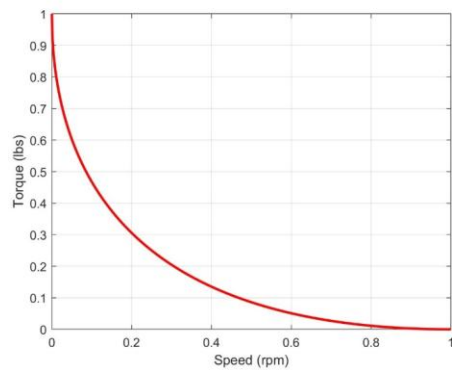


Figure 3. Human's muscle power characteristics

1.2 Previous Project Development

1.2.1 Bike Dynamometer

The previous team sensible solutions built a bike dynamometer built around the Lemond RevMaster stationary exercise bike to measure the torque and speed output of the cyclist. They implemented a torquemeter mounted on the flywheel axle that reflects the torque applied at the pedals of the bike by the cyclist. They used an IR sensor and non-reflective strips to act as spokes for the sensor to count the number of pulses that occur over a set time. The data from the sensors were then collected using an NI DAQ and that data was compiled with a LabVIEW Virtual Instrument.

However, the results of the tests showed inaccuracies in the data. As mentioned earlier, the torquemeter measures the applied torque by the cyclist at the pedals, which measures the input torque of the cyclist. When measuring input torque, it is subject to also factoring in the inertia of the flywheel as its being accelerated by the cyclist. This is seen as noise to the data that is desired for us to properly represent a human being's muscles power characteristic. To solve this problem our goal was to measure braking torque as opposed to the input torque so that the inertia of the

flywheel doesn't affect the data collected as much, giving us a more accurate representation of the muscle's power characteristics.

1.2.2 Hand Dynamometer

The hand dynamometer was initially developed by Grayson Witkowski during the Fall 2024 semester [1] and was later modified by the Sensible Solutions team to address the shortcomings of the original design. The device was designed to measure human grip strength by converting the applied grip force into pressure while simultaneously recording the displacement of a piston within a cylinder. These pressure and displacement signals, acquired in the form of voltage, are later converted into force and velocity for analysis.

In its initial version, the system used air as the working fluid and relied on an Arduino–Raspberry Pi setup for data acquisition. Due to the compressibility of air and limitations in the data acquisition system, the team was unable to obtain reliable and consistent results. In Spring 2025, the Sensible Solutions team improved the design by replacing air with distilled water and upgrading the data acquisition system to a NI DAQ, resulting in better measurement accuracy and data quality.

However, despite these improvements, the results were still affected by the user's initial reaction to the sudden latch release, which introduced inconsistencies in the recorded data. The objective of the present work was therefore to implement necessary modifications and perform experimental testing to obtain data that more accurately represents actual human muscle characteristics.

1.3 Project Objective

Our objective for this project is to measure a human being's power characteristics. We will measure this using the two systems, the bike dynamometer and the hand dynamometer. For the bike dynamometer system, we will measure the muscle power characteristics when pedaling with our legs, and when pedaling with our arms to get the power characteristics of our leg muscles and arm muscles. For the hand dynamometer we are measuring a human being's hand muscle's grip strength power characteristics.

2 THEORY

2.1 Bike Dynamometer

The bike dynamometer which is designed around the Sunny Health & Fitness Sf-B901 chain drive stationary bike is used to measure the velocity of the flywheel and braking torque. Braking Torque can be measured by utilizing Newton's third law of reaction forces. Torque Labs designed a braking yoke around the axial of the flywheel as shown in Fig. 4.

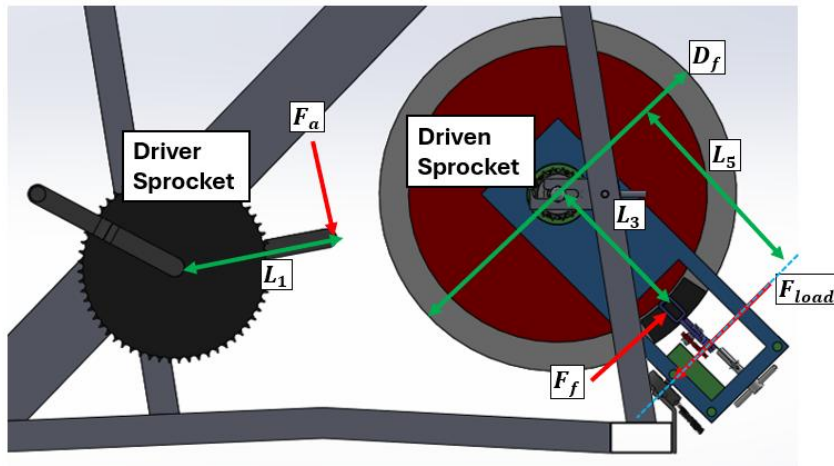


Figure 4. Forces applied on the exercise bike

The load cell will be calibrated by first zeroing it out before the flywheel starts rotating, thus eliminating the weight of the brake yoke. Appendix D shows the entire process on how friction is related to what the load cell will measure. The final equation that will be used to measure braking torque is shown in Eq. 1

$$\tau_{brake} = F_{load} \cdot L_5 \quad (1)$$

The velocity measurement will be done just like the past group Sensible Solutions did. Black vinyl strips with adhesive backing will be placed on the face of the flywheel. An IR sensor will

pulse every time one of these strips passes by. This will ultimately give us the velocity of the flywheel. Figure 5 shows what the flywheel will look like after the vinyl strips are applied.



Figure 5. Previous exercise bike's flywheel with vinyl strips [1]

2.2 Hand Dynamometer

A hand dynamometer operates on the principle of force transmission and fluid flow. When a user applies force on the handle, the force transmitted by the lever arm to a piston within a cylinder causes an change in internal pressure and displacement of the piston. Because the system uses incompressible distilled water, the pressure generated is directly proportional to the applied force. The applied force is determined using moment balance about the pivot. A potentiometer measures the displacement of the piston, which is differentiated with respect to time to obtain velocity. Similarly, a pressure transducer converts the change in pressure into a voltage signal, which is calibrated to obtain the applied grip force. Because the raw signal contains a lot of noise, a zero-phase low-pass filter is applied to analyze the force-velocity relationship of human grip strength. The data measured by this system is then graphed against the theoretical curve defined by the equation:

$$F = F_{max} \left[1 - \left(\frac{V}{V_{max}} \right)^n \right]^{\frac{1}{n}} \quad (2)$$

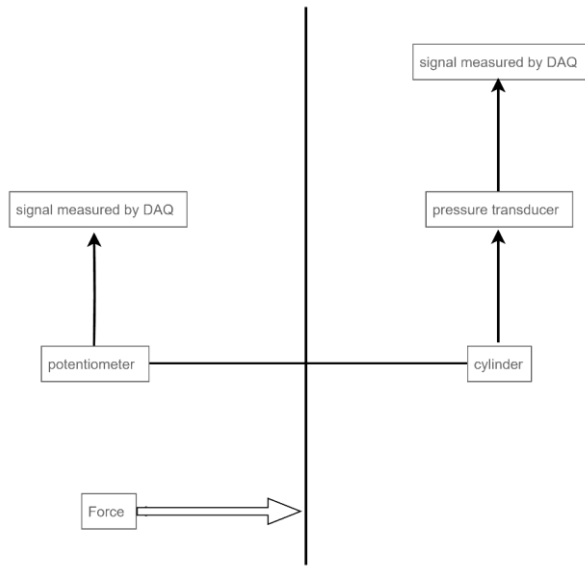


Figure 6. Hand dynamometer layout diagram

3 DESIGN

3.1 Bike Dynamometer

3.1.1 Brake Dynamometer

The exercise bike used for our setup is the Sunny Health and Fitness SF-B901 chain drive stationary bike as seen in Figure 7. Our design for the brake dynamometer consists of a brake yoke that stems from the middle of the flywheel seen in Figure 8.

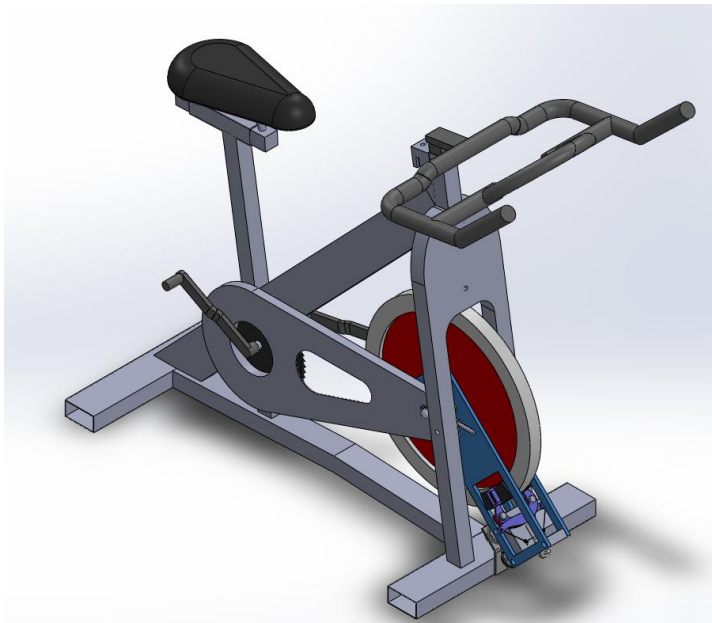


Figure 7. Full bike model

The torque arms sit on bearings on either side, so the flywheels rotation doesn't affect the resulting torque readings on our loadcell, more details about the bearings used in Appendix E.

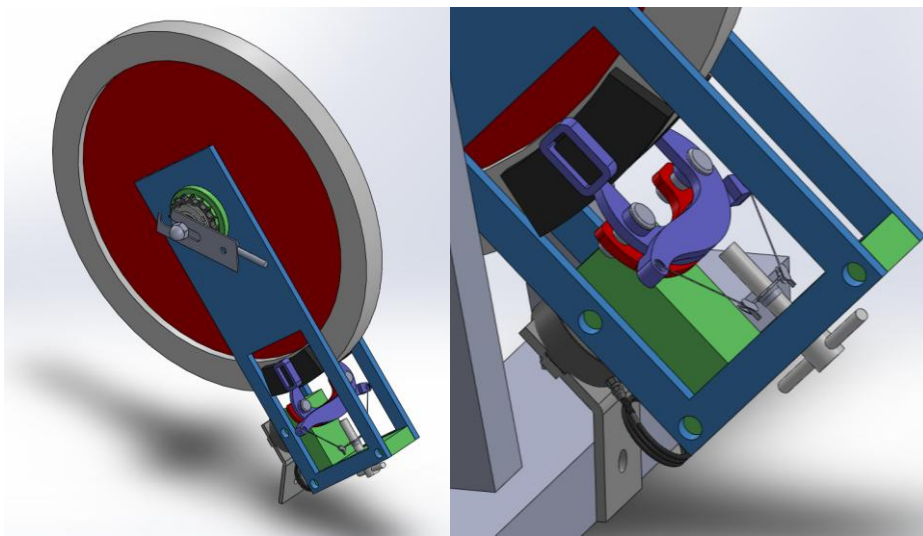


Figure 8. Flywheel with brake dynamometer(left) and close up of brake system (right)

The brakes used for applying resistance to the flywheel are the brakes that come with the exercise bike, the Weinmann center pull brakes, seen in Figure 9. The knob on the brake mount allows us to change the resistance being applied to the flywheel.

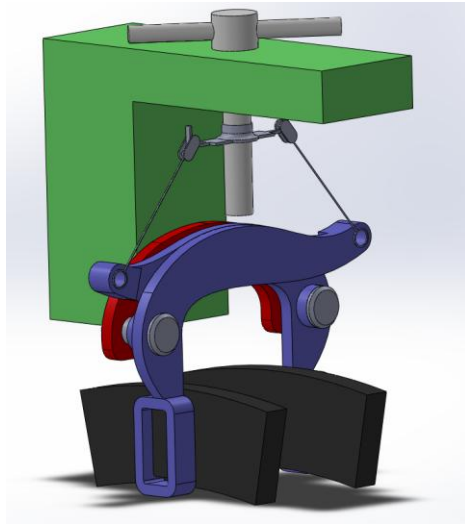


Figure 9. Brake mount and brakes

3.1.2 IR Sensor

The sensor that will be used for measuring the velocity of the flywheel is a reflective IR obstacle avoidance sensor module. This is the same method used by the previous Capstone team, Sensible Solutions. The flywheel will have non-reflective vinyl stripes in the form of spokes. The IR sensor has an IR LED that emits infrared light, the photodiode receives the reflected Infrared light that bounces off the flywheels reflective surface and for every non reflective spoke IR sensor will count that as a pulse. Depending on the number of spokes, the more we have the greater the resolution of the signal will be for the velocity readings. We also needed to consider the mounting of this sensor and the distance from the flywheel itself. Figure 10.

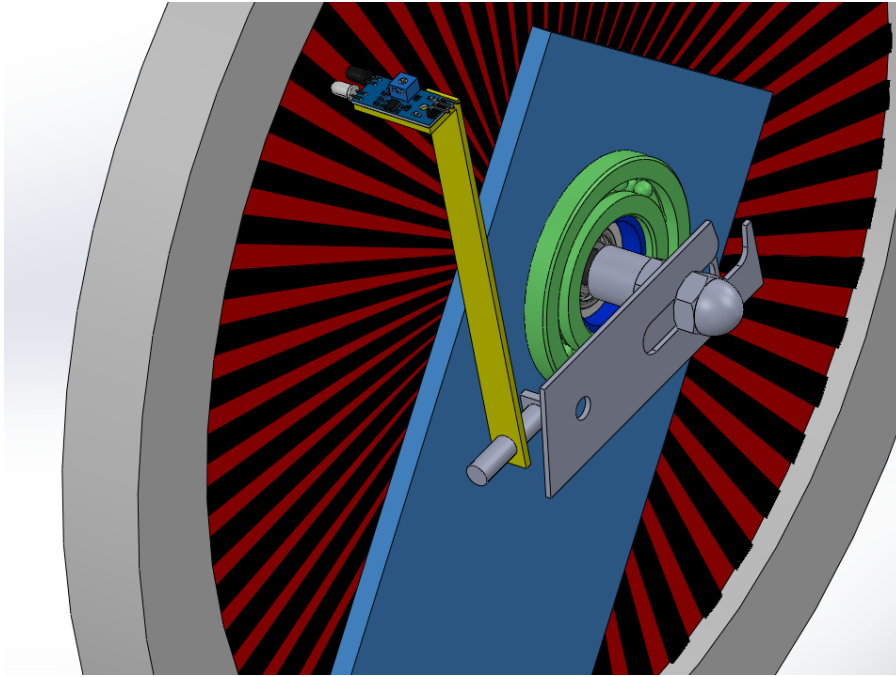


Figure 10. IR Sensor mount on flywheel

3.1.3 Compressive Load Cell

The compressive load cell used to measure the reaction force was sized based on an applied force to the pedal of 180 lbf. This weight was used since an average man is around 180 lbf and it is unlikely for anyone to pedal with a force greater than their body weight. Appendix D shows how the maximum pedal force was used to estimate what the largest force the load cell would have to measure. The derived equation for the friction applied to the flywheel is shown in Eq 3.

$$F_f = \frac{F_a L_1 N_{Driven}}{L_3 N_{Driving}} \quad (3)$$

Using this equation, the max friction produced was estimated to be around 44 lbf. This can be used for sizing the load cell and will ultimately mean the load cell purchased will have a capacity of 100 lbf.

3.2 Hand Dynamometer

The hand dynamometer used in this study was designed and developed by a previous senior design team, and the overall design and configuration were retained, except for the use of different, stiffer material for the locking mechanism. A stiffer locking mechanism was implemented to improve structural integrity and ensure more consistent operation during testing. Therefore, the focus was on improved data acquisition, experimental testing, and analysis.

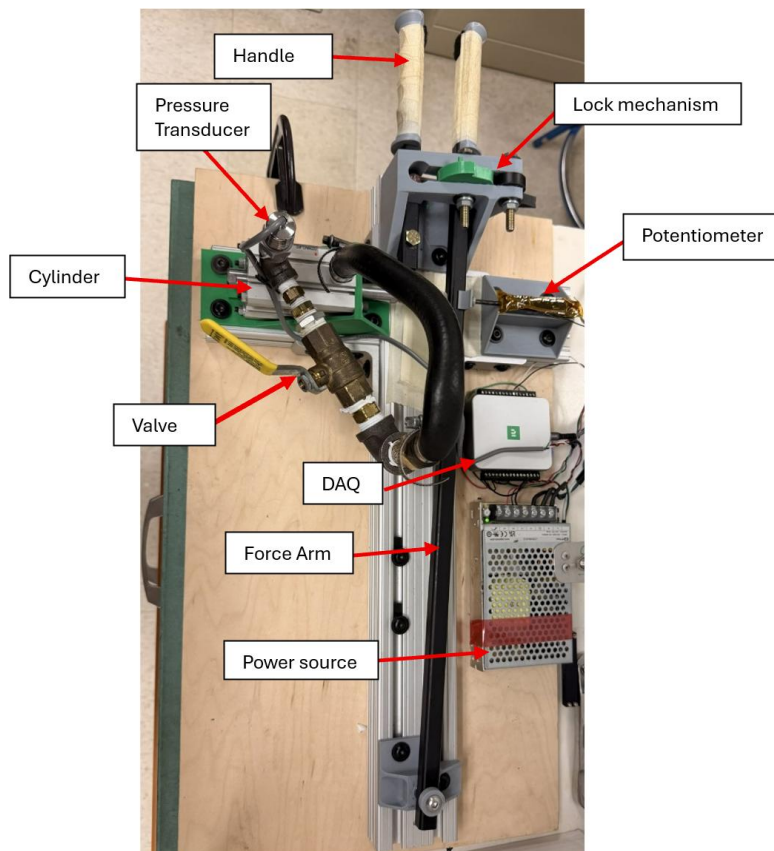


Figure 11. Hand dynamometer present configuration

4 METHODOLOGY

4.1 Bike dynamometer

The experiment test will be conducted using LabVIEW software, where we will begin by starting the program to start data collection. Then the cyclist will begin to pedal the bike and at this time we will be seeing the acceleration from the cyclist, so the program will run until we see the torque output and velocity hit a steady state for 10 seconds at this time we will stop that test run but keep the program on and then increase the resistance applied to the bike. The same process will be repeated from no resistance to a full stalling resistance and that will be one full run for our cyclist.

We are still yet to solidify this method of testing as there are many factors such as fatigue from the cyclist having to pedal against an increasing resistance at full force in back-to-back runs. This can affect the output of the power that the cyclist is able to apply in later runs where the resistance is much higher.

4.2 Hand dynamometer:

The experimental test was conducted using LabVIEW software. To begin, the program was opened and the VI was executed. Initial readings for pressure and displacement were verified to ensure both the potentiometer and pressure transducer were operating to show zero displacement and zero pressure displaying the interface shown in figure 12. While keeping the valve in a fully open position, the handle was actuated once to verify that both sensors were correctly detecting and transmitting signals.

After confirming proper functionality, the valve was adjusted to the desired resistance level. The handle was set to the fully open position, and the locking mechanism was engaged. The program was stopped and restarted to prepare for data collection. The person was asked to use an underhand grip to hold the handle with maximum strength, right after pressing the record button on the LabVIEW interface. Upon activation, the record button on the screen turned neon green, indicating that data acquisition was in progress. Without informing the subject, the locking mechanism was released, allowing the handle to close under the applied grip force.

During the process, the voltage signals displayed on the interface were monitored to ensure the readings changed appropriately. After each trial, the output file generated was checked to confirm successful data collection. The handle was then reset, the valve was set to the next resistance level, and the above procedure was repeated for multiple resistance settings.

For low-resistance trials, a sampling rate of 2000 Hz was used, and data collection occurred automatically once the record button was pressed. For high-resistance trials, a sampling rate of 250 Hz was used, and the record button had to be pressed a second time to terminate data recording and store the collected data.

Initial testing was carried out with multiple people across approximately ten different resistance levels. However, the data showed inconsistencies, mainly because subjects flinched during the

sudden release of the latch, which affected the measurements. To improve the quality and consistency of the data collected, the approach was changed by testing only two people and performing a large number of repeated trials at each resistance level. This allowed the subjects to become more familiar with the latch release and reduced reaction-related variations, resulting in more reliable data.

Each trail produced an excel sheet dataset containing raw voltage signals. The data were trimmed to remove excess data values before latch release and after handle closure. The processed data were then used to compute force and velocity, followed by filtering and extraction of peak values. A detailed procedure is provided in Appendix A.

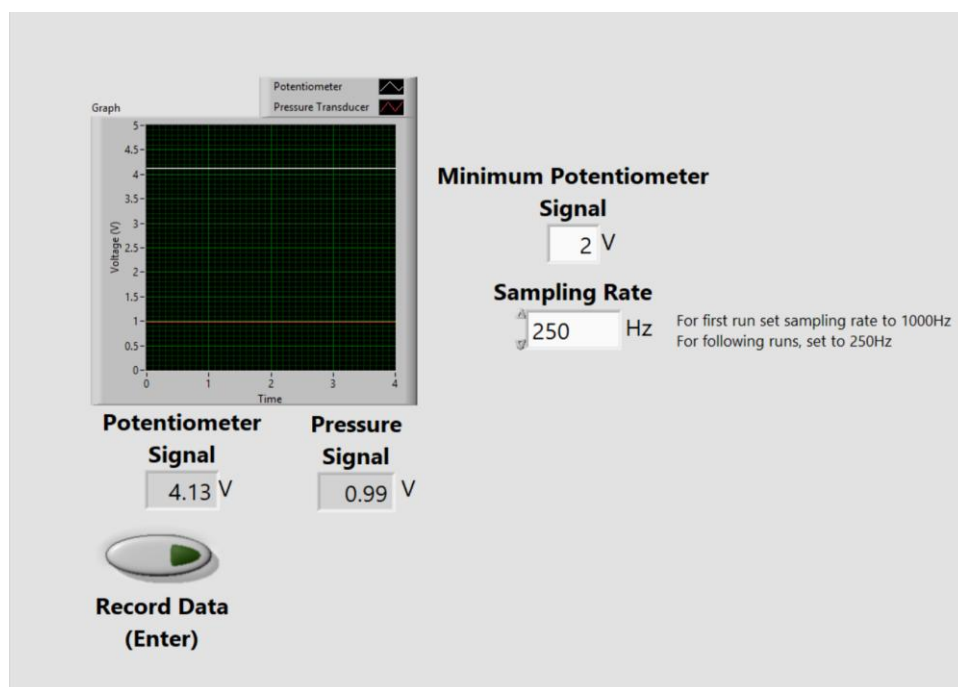


Figure 12. Hand dynamometer LabVIEW interface

5 RESULTS AND DISCUSSION

From the data collected two main Force vs. Velocity graphs were created as shown in Figures 13-14.

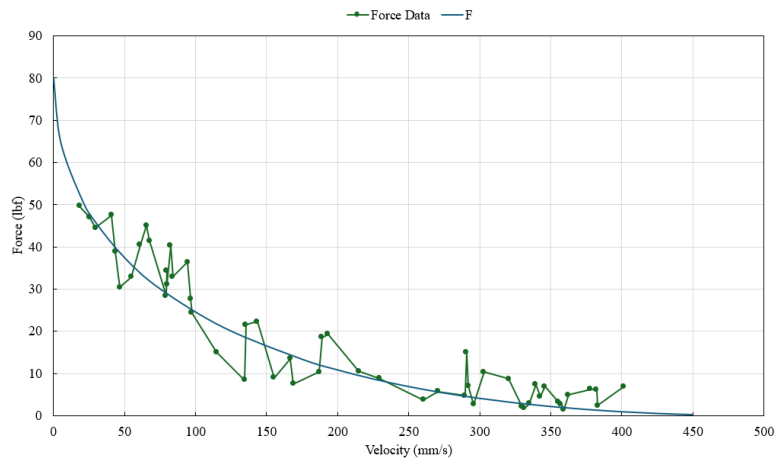


Figure 13. Force vs. Velocity Graph of Participant 1

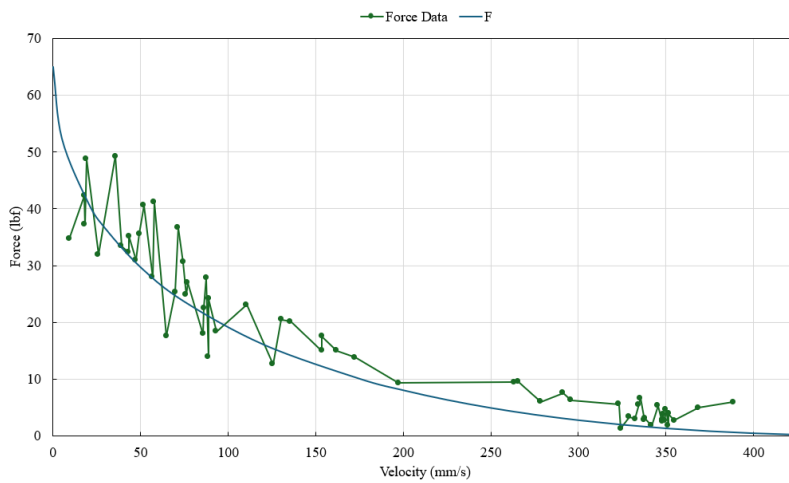


Figure 14. Force vs. Velocity Graph of Participant 2

To analyze the accuracy of the data and theoretical curve, two categories must be considered. First, how much error is contained in the measured values assuming the theoretical values are correct? Second, does the theoretical curve accurately represent the data collected? In order to answer the two questions, three metrics were calculated: the mean absolute error (MAE), found by Equation 4 the mean absolute percentage error (MAPE), found by Equation 5, and the R^2 value, found by Equation 6.

$$MAE = \frac{1}{N} \sum_{i=1}^N |F_{measured} - F_{theoretical}|$$

$$MAE = \frac{1}{N} \sum_{i=1}^N \left| F_{measured} - F_{max} \left[1 - \left(\frac{V}{V_{max}} \right)^n \right]^{\frac{1}{n}} \right| \quad (4)$$

$$MAPE = \frac{100}{N} \sum_{i=1}^N \left| \frac{F_{measured,i} - F_{theoretical,i}}{F_{theoretical,i}} \right|$$

$$MPE = \frac{100}{N} \sum_{i=1}^N \left| \frac{F_{measured,i} - F_{max} \left[1 - \left(\frac{V_i}{V_{max}} \right)^n \right]^{\frac{1}{n}}}{F_{max} \left[1 - \left(\frac{V_i}{V_{max}} \right)^n \right]^{\frac{1}{n}}} \right| \quad (5)$$

$$R^2 = \frac{\sum (F_{measured} - F_{theoretical})^2}{\sum (F_{measured} - \bar{F}_{measured})^2}$$

$$R^2 = \frac{\sum (F_{measured} - F_{theoretical})^2}{\sum \left(F_{measured} - \frac{\sum F_{measured}}{i} \right)^2} \quad (6)$$

The calculated values of the MPE and R^2 are shown in Table 1 below.

Table 1. Analysis Value Results

Test Data Set	MAE (lb)	MAPE (%)	R^2 (%)
Participant 1	4.1	35	88
Participant 2	4.0	34	85

As seen from Table 1 and Figures 13-14, Participant 1 had a slightly larger magnitude of error spread across the data compared to Participant 2. However, the data collected by Participant 1 more closely matched the theoretical value. This would be explained by the extreme outliers found above the curve at low velocities in Figure 13. From the values of R^2 , it can be determined that the collected data generally matches the theoretical curve created by Equation 2.

During the different tests three main sources of error were found to significantly affect the data set. The first being ‘flinching’, at the start of the test signaled by the release of the trigger some participants would loosen their grip as a reaction to the sudden movement. The second being inconsistent grip strength, as seen in the high resistance example data graphs in Appendix #, participants couldn’t always keep a consistent grip. The third being fatigue, after running several tests at high resistances participants lose the strength in their grip. Additionally, some errors may have been introduced during data processing, as the zero-phase filtering could have been improved by selecting a more appropriate T/τ value for high and low resistance runs.

To consider how changing the value of n in Equation 2 would affect the theoretical curve’s correlation to the measured data, Table 2 lists different values of n and the corresponding value of R^2 .

Table 2. n Considerations

N	Participant 1 R ² (%)	Participant 2 R ² (%)
0.60	68	75
0.59	72	78
0.58	76	81
0.57	80	83
0.56	83	85
0.55	85	86
0.54	87	87
0.53	88	87
0.52	89	87
0.51	89	86
0.50	88	85
0.49	86	83
0.48	84	80
0.47	81	77
0.46	77	73
0.45	72	69
0.44	67	63
0.43	61	57
0.42	54	50
0.41	47	43
0.40	38	35

Commented [MV1]: Is it possible to make these tables side by side so that Table 3 does get cut off

From this table, $n = 0.5$ should be the correct value for the theoretical equation. The highest values of n can be found at 0.52 on Table 2; however, this can be explained by error and not a different value of n as it is not significant enough of a difference in R^2 to suggest that the value of n is inaccurate.

6 CONCLUSIONS

The hand dynamometer experiment successfully demonstrated that human muscle characteristics can be measured with the combination of mechanical components, sensors, data acquisition, and data analysis. By converting input force into pressure and measuring the displacement inside the cylinder in the form of electrical signals, the device allowed us to determine the force–velocity relationship of human grip strength. The use of a stiffer lock and multiple trials per person helped us to collect reliable data. Despite these improvements, a few other sources of error were present, including flinching, inconsistent grip strength, and fatigue. In addition to those, data analysis choices, such as cutoff frequency for velocity and force, influenced the outcome. Overall, the experiment provided valuable outcomes and insight into the force–velocity relationship, and further improvements are necessary in the areas of design and data analysis. This test data leads to the affirmation that n should be equal to 0.5 in the theoretical equation.

7 FUTURE WORK

7.1 Bike Dynamometer

Our goal for the next semester is to have the bike and data acquisition method built and set up for testing. We will also be finalizing our testing methodology, our main criteria is going to be consistence and accuracy of the runs and maintaining a consistent torque output from the cyclist pedaling with their arms and legs while the resistance is being increased every increment will be a challenge due to fatigue from the cyclist and we will create a solutions for that in our testing methodology. We will be going thorough calibrating the IR sensor and load cell and initially finding the ideal run time of each test run at the different resistance levels. Finding the ideal sampling frequency and number of spokes that we should be running for our velocity sensor to get a good resolution for our data. Once we have data for our arms and legs it will give us a more accurate representation of a human being's muscles power characteristics paired with the data already acquired form the hand dynamometer.

7.2 Hand Dynamometer

The reliability of the hand dynamometer can be further improved in several ways. One important next step is to improve the lock system to minimize the initial noise that occurs at the moment of latch release. This would increase the quality of the data measured by the device. Future work should also focus on data processing, primarily the selection of filtering parameters. Since cutoff frequency and sampling frequency affect the value of T/τ , and thus the degree of smoothing, a systematic way to choose them would be beneficial.

Increasing the sample size and diversity of test subjects is another important step to better represent the range of human muscle characteristics. This would help to make meaningful comparisons among different individuals.

8 ACKNOWLEDGEMENTS & REFERENCES

Dr. Robert Woods – Faculty Advisor

John Skiba – Graduate Student, former member of the Hand Dynamometer group

References

- [1] Witkowski, Grayson, 2024, “Hand Dynamometer,” The University of Texas at
Arlington, Arlington, TX

9 NOMENCLATURE

Greek Symbols:

τ Torque

θ Angle

Roman Symbols:

D Diameter

F Force

f Value determined by an Equation

G Number of gear teeth

L Length

n Theoretical Curve Power

N Total Number of Data points

R Reaction Force

R^2 Coefficient of Determination

V Velocity

W Weight

X Position

P Pressure

Subscripts:

$n+1$ Next Term

$n-1$ Last Term

10 APPENDIX A: DATA PROCESSING

For all test data collected two sheets are created in Excel file, one containing the raw data and the other containing a cut version of the raw data. The cut version should include only the data of the run of interest for analysis. The cut data should start at constant pressure voltage for at least five data points before the pressure increases significantly signaling the start of a test. An example of how the data should be cut can be found in Table 3.

Table 3. Top Cut of Data

Position (V)	Pressure (V)
4.101025	1.007579
4.101025	1.008546
4.101348	1.008224
4.101025	1.009514
4.101025	1.009514
4.100703	1.009836
4.100703	1.010158
4.100381	1.010481
4.100381	1.01177
4.099736	1.01306
4.096835	1.024342
4.071368	1.188106
4.047835	2.203888
4.014308	3.730624

More data points can be taken away from the initial increase of pressure, but at least five need to be taken at a steady pressure reading before the test begins. For the bottom cut of the raw data, only the data point where the test has ended needs to be recorded. This will be determined when the potentiometer reads a voltage between 1.9 - 2.0. More points after the test can be saved but there is no need as those specific data points hold no relevance other than signifying the end of a test. An example of the bottom cut of the data can be found in Table 4.

Table 4. Bottom Cut of Data

Position (V)	Pressure (V)
2.077197	1.178112
2.066237	1.167474
2.056243	1.156836
2.04625	1.145875
2.036901	1.131691
2.027875	1.114606

2.019816	1.096875
2.011112	1.080112
2.00273	1.063027
1.995638	1.049487
1.987901	1.034981
1.980809	1.022731
1.974039	1.010481
1.966947	1.003066

The cut data should then be copied into the zero-phase filter sheet as shown in Figure 14. Through experimentation it was found that .95 was the best (insert) constant for the filter.

	A	B	C	D	E	F	G	H	I	J	K	L
1		T/tau =	0.95		T/tau =	0.95						
2		exp =	0.2867		exp =	0.2867						
3		1-exp =	0.6133		1-exp =	0.6133						
4												
5		Insert Data:										
6			X (in)	F (in)		Backward		Forward		Filtered		
7			4.1133	1.0321		4.1133	1.0321	4.1127	1.0328	4.113	1.0324	
8			4.1129	1.0305		4.1127	1.0311	4.1119	1.0336	4.1123	1.0325	
9			4.1126	1.0321		4.1127	1.0317	4.1113	1.0352	4.112	1.0354	
10			4.112	1.0324		4.1122	1.0321	4.1091	1.0355	4.1107	1.0413	
11			4.1101	1.0359		4.1109	1.0345	4.1044	1.0792	4.1077	1.0568	
12			4.1049	1.0485		4.1072	1.0431	4.0851	1.1477	4.1014	1.0964	
13			4.0888	1.1185		4.0959	1.0893	4.0807	1.3051	4.0883	1.1072	
14			4.0759	1.3267		4.0836	1.2349	4.0679	1.6009	4.0758	1.4179	
15			4.062	1.7726		4.0784	1.6646	4.0552	2.0377	4.0628	1.8002	
16			4.0491	2.5537		4.0573	2.1712	4.0444	2.4531	4.0509	2.3121	
17			4.0433	2.8813		4.0487	2.484	4.0389	2.2936	4.0428	2.3888	
18			4.032	3.8967		4.0385	2.1238	4.0295	1.6788	4.0325	1.9013	
19			4.0249	4.1889		4.0302	1.6915	4.018	1.3333	4.0241	1.5124	
20			4.014	4.2207		4.0203	1.4028	4.0059	1.1975	4.0136	1.3001	
21			4.0037	4.1559		4.0101	1.214	3.9959	1.1668	4.0028	1.2001	
22			3.9927	4.1475		3.9994	1.1877	3.9827	1.1687	3.9911	1.1762	
23			3.9776	4.1697		3.985	1.1767	3.9699	1.2023	3.9765	1.1895	
24			3.9602	4.2148		3.9702	1.2021	3.9501	1.2539	3.9601	1.227	
25			3.9444	4.2855		3.9543	1.2524	3.9341	1.3159	3.9442	1.2842	
26			3.9273	4.3532		3.9377	1.3142	3.9178	1.3842	3.9278	1.3392	
27			3.9131	4.3857		3.9226	1.3661	3.9028	1.3818	3.9127	1.3699	
28			3.8966	4.3851		3.9067	1.3746	3.8865	1.3755	3.8966	1.3751	
29			3.8802	4.3644		3.8904	1.3684	3.8704	1.3603	3.8804	1.3643	
30			3.8654	4.347		3.8751	1.3553	3.8548	1.3538	3.8649	1.3545	
31			3.8483	4.3519		3.8586	1.3532	3.838	1.3644	3.8483	1.3588	
32			3.8318	4.377		3.8422	1.3678	3.8216	1.3843	3.8319	1.3761	
33			3.8164	4.4009		3.8264	1.3881	3.8059	1.3959	3.8159	1.382	
34			3.7993	4.408		3.8098	1.4003	3.7882	1.3881	3.799	1.3942	
35			3.7809	4.388		3.7921	1.3927	3.7705	1.3955	3.7813	1.3746	
36			3.7642	4.3438		3.7749	1.3621	3.754	1.3966	3.7645	1.3344	
37			3.748	4.2829		3.7584	1.3135	3.7378	1.2491	3.7481	1.2813	
38			3.7319	4.2187		3.7422	1.256	3.7215	1.1956	3.7319	1.2258	
39			3.7155	4.1572		3.7256	1.2015	3.7051	1.1573	3.7154	1.1794	
40			3.6984	4.1336		3.709	1.1599	3.6888	1.1417	3.6988	1.1508	
41			3.6802	4.1278		3.6932	1.1402	3.6731	1.1544	3.6831	1.1473	
42			3.6674	4.153		3.6774	1.1448	3.657	1.1955	3.6672	1.1723	
43			3.6513	4.211		3.6614	1.1866	3.6405	1.2855	3.6509	1.2261	
44			3.6342	4.3019		3.6447	1.2573	3.6233	1.352	3.634	1.3047	

Figure 15. Excel Zero Phase Filter

Then convert the filtered data into the position and force measurements as shown in Appendix C. The test data will generally converge into three types of shapes, one for low resistance tests and two for high resistance tests. For low resistance tests, the data point to be taken should occur after the dynamic effect, but before the velocity drops off. Note that for both high and low resistance test data the data point should be taken at a generally constant velocity. The considerations for where to take the data point for low resistance test data can be seen in Figure 16.

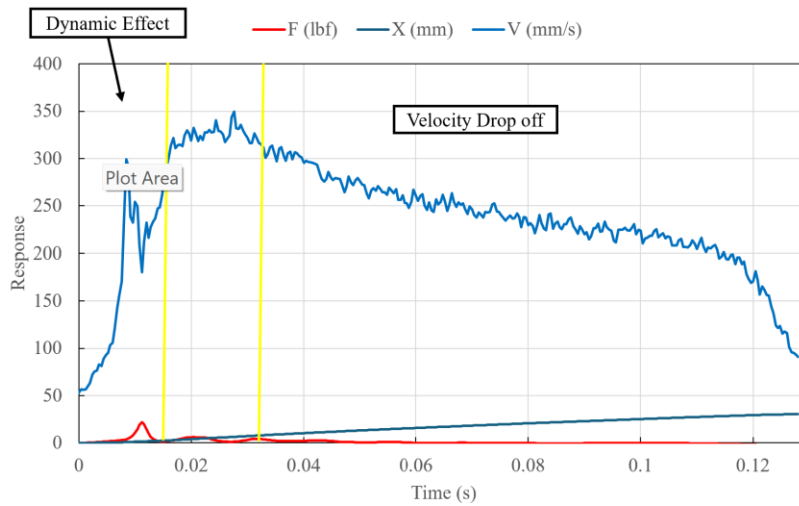


Figure 16. Low Resistance Example Data Graph

The shape of a high resistance run will have a flat velocity and force curve after the dynamic effect or one with a single hump, and the velocity decreasing at the end like in the low resistance tests. The data point for graphs of these characteristics should be anywhere at constant velocity or at the peak of the hump but before velocity drops off as seen in Figures 17-18.

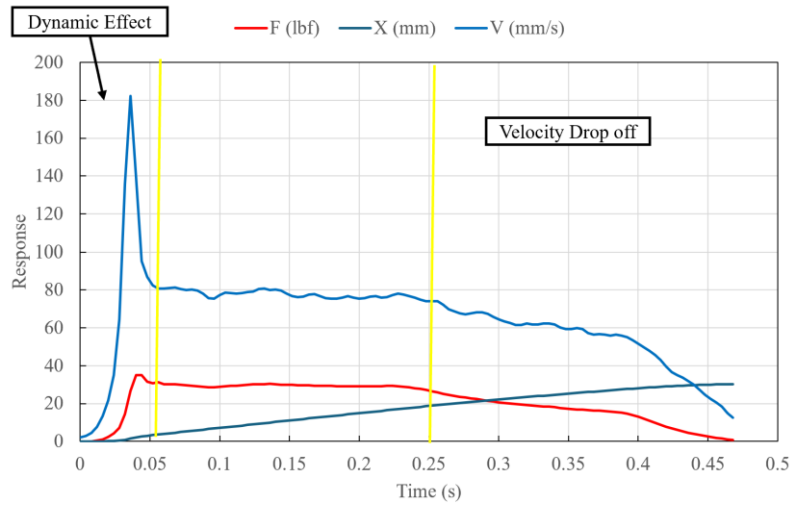


Figure 17. High Resistance Example Data Graph 1

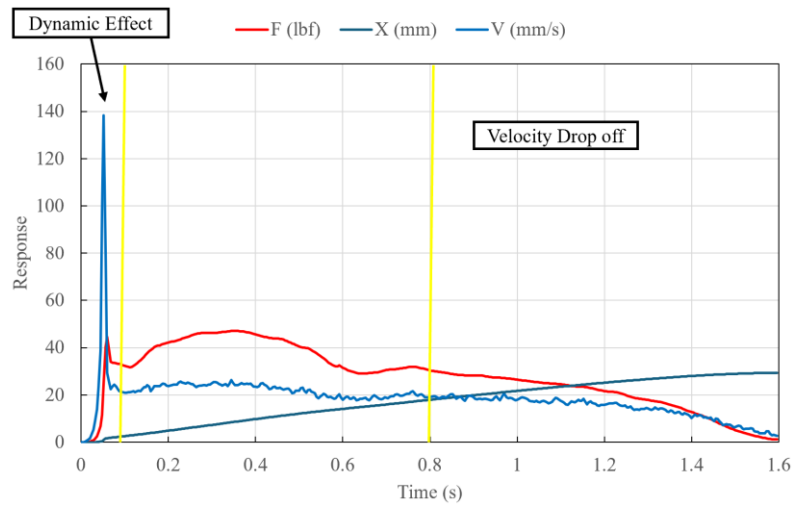


Figure 18. High Resistance Example Data Graph 2

Each individual data point from every test should then be placed in a Force vs. Velocity table to be graphed. Along with the data point from a collective of tests, the Force vs. Velocity graph as seen in Figure 19 also contains the theoretical curve found by Equation 2 where n is equal to 0.5:

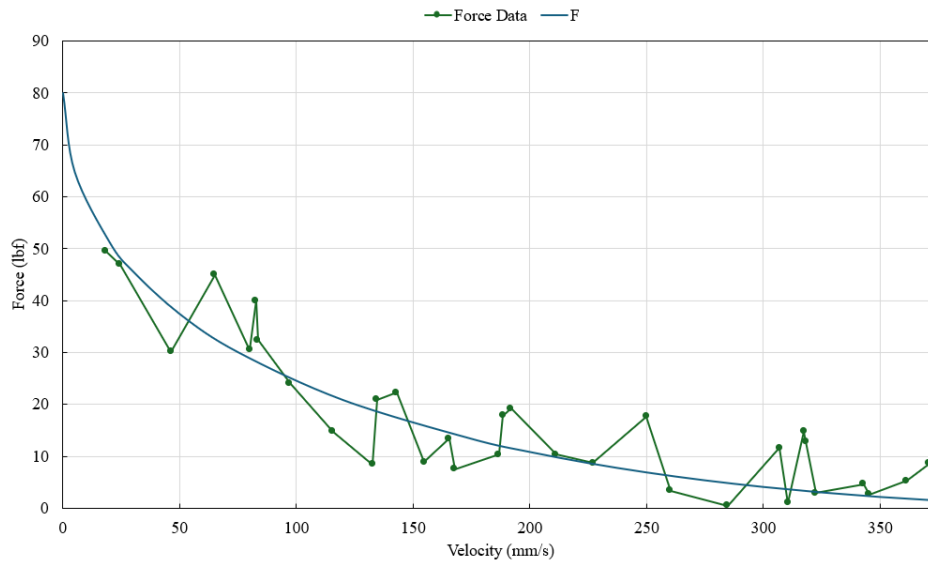


Figure 19. Force vs. Velocity Graph Example

The values used for the maximum force should be the recorded maximum force of the participant on the (insert device name here). The maximum velocity used in the formula should typically be around 500 mm/s but can be lowered to around 450 mm/s depending on the participant.

11 APPENDIX B: CALIBRATION

CALIBRATION EQUATIONS FOR HAND DYNAMOMETER

Displacement Calibration

$$X(\text{mm}) = 18.4 (4.10 - V_x)$$

Where:

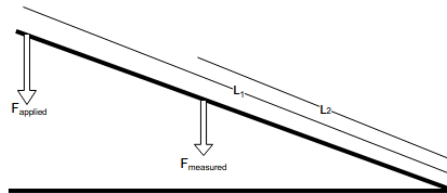
$X(\text{mm})$ = displacement in millimeters

V_x = measured position voltage (V)

4.10 V = zero-displacement reference voltage

18.4 mm/V = displacement calibration factor

Force Calibration



Knowns:

- $A = 1.62 \text{ in}^2$ (area of piston)
- $L_2 = 17 \text{ in}$
- $L_1 = 24.5 \text{ in}$

$$F_{\text{applied}} = \frac{L_1}{L_2} \cdot F_{\text{measured}}$$

$$F_{\text{measured}} = P \cdot A$$

$$F_{\text{applied}} = \frac{L_1}{L_2} \cdot (P \cdot A)$$

$$\begin{aligned}\frac{L_1}{L_2} &= \frac{24.5}{17} = 1.441 \\ F_{\text{applied}} &= 1.441 \cdot (P \cdot 1.62) \\ F_{\text{applied}} &= 2.34 \cdot P\end{aligned}$$

$$\boxed{F_{\text{applied}} = 2.34 P}$$

where:

- P is in **psi**
- F_{applied} is in **lbf**

Including Voltage Calibration

$$\begin{aligned}P &= 25(V_p - 1) \\ F_{\text{applied}} &= 2.34 \cdot 25(V_p - 1)\end{aligned}$$

$$\boxed{F_{\text{applied}} = 58.5(V_p - 1)}$$

Where:

F_{applied} = applied force by the user (lbf)

F_{measured} = force generated by fluid pressure (lbf)

P = pressure inside the cylinder (psi)

A = piston cross-sectional area (in²)

L_1 = distance from pivot to applied force location (in)

L_2 = distance from pivot to measured force location (in)

V_p = measured pressure voltage (V)

1 V = zero-pressure reference voltage

25 psi/V = pressure calibration factor

First-Order Low Pass Filter

Continuous Form:

$$\tau \frac{dy}{dt} + y(t) = x(t)$$

Transfer Function:

$$G(s) = \frac{1}{\tau s + 1}$$

Cutoff Frequency:

$$\omega_c = \frac{1}{\tau}$$
$$f_c = \frac{1}{2\pi\tau}$$

Sampling Frequency:

$$f \text{ (Hz)}, T = \frac{1}{f}$$

Discrete Implementation:

$$y_k = y_{k-1} + (1 - e^{-T/\tau})(x_k - y_{k-1})$$

Backward Difference

$$f(x_k - x_{k-1})$$

Forward Difference

$$f(x_{k+1} - x_k)$$

Central Difference

$$\frac{f}{2}(x_{k+1} - x_{k-1})$$

Filtered Velocity Equation

$$V_k = V_{k-1} + \left(1 - e^{-\frac{1}{T\tau}}\right)(V_{\text{raw},k} - V_{k-1})$$

HAND DYNAMOMETER CYLINDER:

Parameters

$$\begin{aligned} D_{\text{cylinder}} &= 45 \text{ mm} \\ D_{\text{piston rod}} &= 16 \text{ mm} \\ L_{\text{stroke}} &= 40 \text{ mm} \\ L_{\text{displacement}} &= 20 \text{ mm} \\ T_{\text{stroke}} &= 0.15 \text{ s (ideal time for zero resistance)} \end{aligned}$$

Piston Area

$$\begin{aligned} A_{\text{piston}} &= \frac{\pi}{4}(D_{\text{cylinder}}^2 - D_{\text{piston rod}}^2) \\ A_{\text{piston}} &= \frac{\pi}{4}(45^2 - 16^2) \\ A_{\text{piston}} &= 1389.4 \text{ mm}^2 = 2.15 \text{ in}^2 \end{aligned}$$

Displacement Volume

$$\begin{aligned} V_{\text{displacement}} &= A_{\text{piston}} \times L_{\text{displacement}} \\ V_{\text{displacement}} &= 27,787.4 \text{ mm}^3 = 1.70 \text{ in}^3 \end{aligned}$$

12 APPENDIX C: EXCEL DOCUMENT SETUP

Each Excel document contains four sheets to analyze individual tests: Raw Data, Cut Data, Auto-Filter, and Auto Graph. The Raw Data sheet is the data exported by LabVIEW in three columns: Time, Voltage_0 (Position), and Voltage_1 (Pressure). The Cut Data sheet is just a page to contain the trimmed version of the raw data to preserve all of the original data, but also to remove the data points that serve no purpose. An example of the Raw Data sheet can be seen in the figure below.

	A	B	C	D	E	F	G	H
1	Time	Voltage_0	Voltage_1					
2	00.000	4.133262	1.006612					
3	00.001	4.132296	1.00629					
4	00.002	4.132617	1.006612					
5	00.003	4.132617	1.006612					
6	00.004	4.133262	1.006612					
7	00.005	4.133262	1.006612					
8	00.006	4.133262	1.005967					
9	00.007	4.13294	1.006612					
10	00.008	4.13294	1.005645					
11	00.009	4.133262	1.00629					
12	00.010	4.13294	1.005967					
13	00.011	4.133262	1.005967					
14	00.012	4.133262	1.005945					
15	00.013	4.13294	1.00629					
16	00.014	4.132617	1.005645					
17	00.015	4.133262	1.006035					
18	00.016	4.132617	1.00629					
19	00.017	4.13294	1.006612					
20	00.018	4.13294	1.005967					
21	00.019	4.133262	1.00629					
22	00.020	4.133262	1.005967					
23	00.021	4.133262	1.006612					
24	00.022	4.133262	1.00629					
25	00.023	4.132617	1.005967					
26	00.024	4.13294	1.00629					
27	00.025	4.13294	1.005967					
28	00.026	4.133262	1.00629					
29	00.027	4.133262	1.006612					
30	00.028	4.133262	1.00629					
31	00.029	4.13294	1.006035					
32	00.030	4.133585	1.00629					
33	00.031	4.133262	1.00629					
34	00.032	4.13294	1.00629					
35	00.033	4.132617	1.006035					
36	00.034	4.133585	1.00629					
37	00.035	4.133262	1.006035					

Figure 20. Example of Raw Data Sheet in Excel

The Auto-Filter sheet is where the cut data is then filtered by the zero-phase filter as shown in Appendix A. For the zero-phase filter, the user inputs the two exponential value labeled T/tau that is calculated below as $1 - \exp(-T/\tau)$. There is an exponential value for the position and pressure voltage data. For the Backward section of the zero phase filter the first value is defined by the formula $=IF(ISNUMBER(B7),B7,"")$. This formula detects if there is a value in the inserted data column and then assigns the cell the same value as the number in the inserted data column. All values after the first in the Backward column have the formula $=IF(ISNUMBER(B8),E7* \$B\$2 + B8* \$B\$3, "")$ with increasing row values. This formula detects if there is a value in the Insert Data column in the same row, and if so, it calculates:

$$f = x_{n-1} \left(e^{T/\tau} \right) + x_n \left(1 - e^{T/\tau} \right)$$

The Forward Column is calculated by the formula =IF(AND(ISNUMBER(B7), ISBLANK(B8)), B7, IF(AND(ISNUMBER(B8), ISNUMBER(B7)), \$B\$2*H8+\$B\$3*B7, "")), where the row values increase. This formula calculates where the last row in the Insert Data column contains a value and then assigns that value to the cell in the Forward column. If the row isn't the last row, it calculates the formula:

$$f = x_{n+1} \left(e^{T/\tau} \right) + x_n \left(1 - e^{T/\tau} \right)$$

The Auto-Graph Sheet is shown in Figure # below.

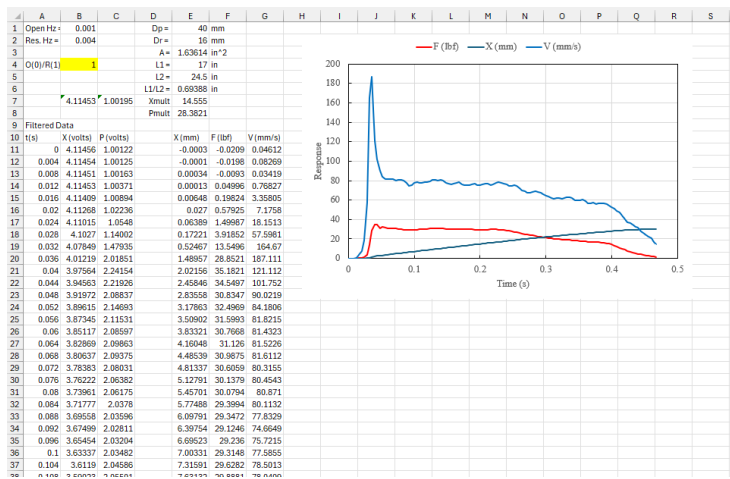


Figure 21. Auto-Graph Sheet

Like in the Auto-Filter Sheet, the X(mm), F(lbf), and V(mm/s) are set to have an if condition to only initiate the cells formulas if there is a corresponding data point in the X(volts) and P(volts) on the same row. To convert the voltages into mm and lbf, some multiples were created at the top of the page. Xmult contains the equation $=2*25.4/5.03/(L1/L2)$, and Pmult contains the equation $=(100/4)*A*(L1/L2)$. The reasons Xmult and Pmult are assigned to these equations can be found in the math of Appendix B. The X(mm) cells are calculated by the formula $(X_{avg}-X(volts))*Xmult$, where Xavg is the average of the first five X(volts) values. The P(lbf) cells are calculated by the formula $(P(volts)-P_{avg})*Pmult$, where Pavg is the average of the first five P(volts) values. The Velocity column uses the equation $V(n) = \frac{X_{n+1}-X_{n-1}}{2(\frac{1}{frequency})}$. At the first and last

rows of the velocity calculation the formula only uses the current position minus the previous or next position divided by the change in time.

13 APPENDIX D: LOAD CELL SIZING/BRAKE TORQUE MEASUREMENT

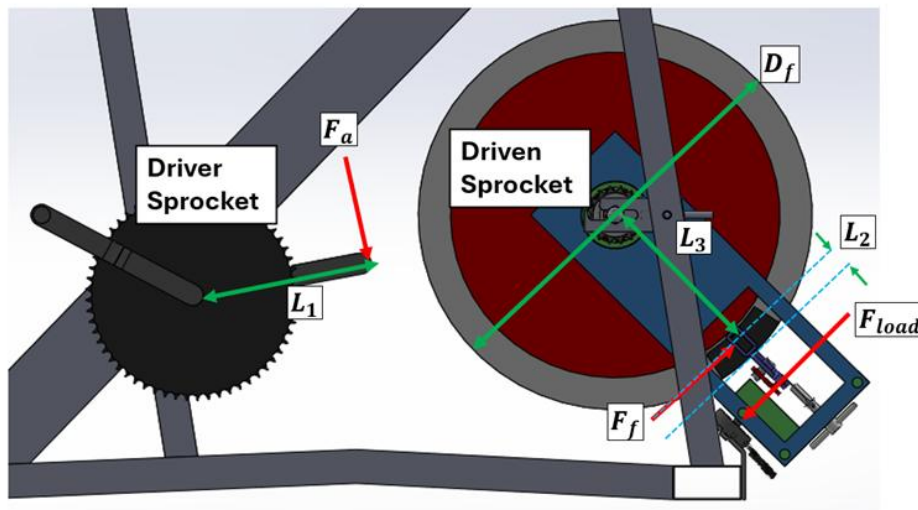


Figure 22. Forces applied on bike

Assumptions:

- 1) No frictional losses between chain and sprockets
- 2) Symmetry is assumed
- 3) Bike is stalled

The following parameters were measured and used for calculation:

Parameters	Description	Quantity	Units
Ndriving	Number of Gears of Driving Sprocket	52	unitless
Ndriven	Number of Gears of Driven Sprocket	16	unitless
Df	Diameter of Flywheel	18	in
L1	Length of Pedal	6 5/8	in
L2	Length from Friction Force to Edge of Flywheel	5/8	in
L3	Length of Friction Force to Center of Flywheel	8 3/8	in
Fa	Applied Force to Pedal	180	lbf

Figure 23. Parameters for loadcell sizing

Starting with the Gear Ratio:

$$Gear\ Ratio = \frac{N_{Driving}}{N_{Driven}} = \frac{\tau_a}{\tau_{out}}$$

Static equilibrium of driven sprocket and flywheel gives:

$$\tau_{out} = F_f L_3$$

The applied torque is equal to:

$$\tau_a = F_a L_1$$

Combining equations gives:

$$\frac{G_{Driving}}{G_{Driven}} = \frac{F_a L_1}{F_f L_3}$$

$$F_f L_3 \frac{G_{Driving}}{G_{Driven}} = F_a L_1$$

Solving for friction gives

$$F_f = \frac{F_a L_1 G_{Driven}}{L_3 G_{Driving}}$$

The Figure below is a FBD of one of the arms of the braking yoke.

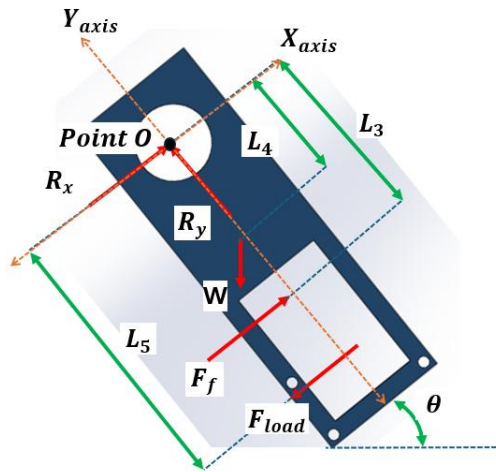


Figure 24. FBD of on one arm of the braking yoke

Assumptions:

- 1) The brake yoke is rigidly attached to the center piece
- 2) Symmetry is assumed on either side of the Flywheel
- 3) The weight represents the weight of entire brake yoke
- 4) The brake is in static equilibrium

Equations

$$\sum F_x = 0 = R_x + F_f - F_{load}$$

$$\sum F_y = 0 = -W \sin \theta + R_y$$

$$\sum M_o = 0 = -W \cos \theta \cdot L_4 + F_f \cdot L_3 - F_{load} \cdot L_5$$

Combining Equations

$$F_{load} \cdot L_5 = -W \cos \theta \cdot L_4 + F_f \cdot L_3$$

$$F_{load} = \frac{F_f \cdot L_3}{L_5} - \frac{W \cos \theta \cdot L_4}{L_5}$$

Braking Torque can then be calculated as:

$$\tau_{brake} = F_{load} \cdot L_5$$

To calibrate the load cell before the flywheel rotates, the load cell will be zeroed out eliminating the weight term in the equation. The lengths were estimated based on the CAD of the bike. To estimate the max friction force to be applied, 180 applied forces to pedal were used.

$$F_f = \frac{(180 lbf)(6 \frac{5}{8} in)(16)}{(8 \frac{3}{8} in)(52)}$$

$$F_f = 44 lbf$$

$$F_{load} = \frac{(44 lbf)(8 \frac{3}{8} in)}{(10 \frac{1}{2} in)} \approx 35 lbf$$

14 APPENDIX E: BEARING SELECTION

Must haves:

- 1) Handle radial load
- 2) Have a thickness less than 22 mm
- 3) 40 mm inner bore diameter

Bearing Decision Matrix							
Criterion	Weight	Deep groove ball bearings		Angular contact ball bearings		Self Aligning Ball Bearings	
Visual							
Sensitivity to Misalignment	40%	2	0.8	2	0.8	5	2
Low Friction	40%	5	2	4	1.6	5	2
Cost	20%	5	1	4	0.8	4	0.8
Total Score			3.8		3.2		4.8

Figure 25. Bearing Decision Matrix

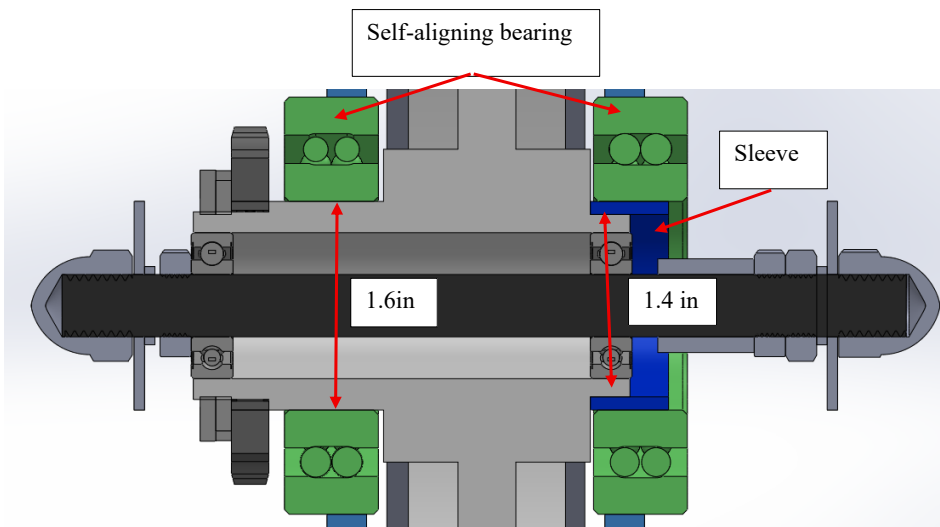


Figure 26. Cross section view flywheel axle

15 APPENDIX F: EXAMPLE DATA

Below is a table containing a cut data set from one of the tests.

Table 5. Cut Data Set Example

X(volts)	P(volts)
4.114565	1.001132
4.114565	1.00081
4.114565	1.000165
4.114887	1.000487
4.114887	1.00081
4.114243	1.002099
4.114887	1.001132
4.114565	1.002099
4.097157	1.394421
4.001736	2.169072
3.970144	2.390217
3.94242	2.282868
3.91792	2.023039
3.895354	2.173908
3.873111	2.116204
3.85119	2.068816
3.828624	2.103954

3.806381	2.098796
3.783493	2.08171
3.762539	2.059145
3.739328	2.068816
3.71773	2.029487
3.694841	2.035934
3.674855	2.022717
3.654868	2.030776
3.633591	2.031099
3.611993	2.046572
3.590072	2.058178
3.569118	2.054632
3.547519	2.062691
3.52592	2.065592
3.504644	2.091382
3.482401	2.080743
3.459835	2.089447
3.437914	2.108467
3.415993	2.086546
3.392782	2.082033
3.371828	2.072362
3.35023	2.074941
3.329598	2.068816
3.308967	2.064947
3.287368	2.073329
3.265769	2.056888
3.243848	2.054954
3.223861	2.059145
3.202585	2.051408
3.182276	2.05173
3.161644	2.054632
3.140046	2.037546
3.119092	2.051408
3.098782	2.036579
3.077828	2.044638
3.057197	2.050441
3.035276	2.037224
3.014967	2.061724
2.994335	2.061401
2.973059	2.063658
2.95146	2.046895
2.929539	2.044638

2.908585	2.030454
2.886986	2.020138
2.866677	2.020783
2.846046	1.98371
2.826059	1.958566
2.80446	1.949217
2.784151	1.908921
2.765131	1.895382
2.745789	1.878618
2.727092	1.856697
2.709039	1.845414
2.690342	1.830908
2.671644	1.82575
2.652302	1.798026
2.633605	1.781263
2.614908	1.765145
2.597177	1.752895
2.579447	1.732263
2.562039	1.728395
2.545276	1.714211
2.528835	1.716467
2.511105	1.697447
2.494342	1.686165
2.477579	1.683586
2.460171	1.676816
2.443085	1.674559
2.425355	1.662309
2.408914	1.644579
2.392796	1.642645
2.376355	1.622336
2.359914	1.621369
2.342506	1.615244
2.327033	1.604928
2.311881	1.595901
2.295763	1.601059
2.280289	1.589776
2.265138	1.588487
2.249342	1.574948
2.233546	1.57527
2.218072	1.558829
2.202599	1.532395
2.188092	1.506605

2.173585	1.462119
2.160046	1.427625
2.146506	1.381849
2.133612	1.345421
2.122329	1.30577
2.111691	1.271277
2.102342	1.241941
2.091704	1.216474
2.083322	1.192619
2.073651	1.168763
2.065592	1.143619
2.0585	1.1275
2.051408	1.110092
2.045605	1.091073
2.038835	1.074632
2.033355	1.059158
<u>2.028842</u>	<u>1.046908</u>

The figure below is the graph generated by the cut data.

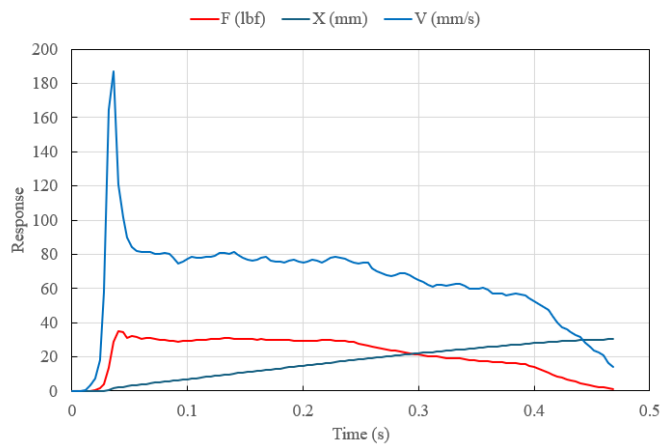


Figure 27. Example Graph

More data from tests is stored on flash drive and is available upon request.